

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 10

SWAPPING OF TWO NUMBERS

AIM: To write and execute an 8085 assembly language program for swapping of two numbers two numbers and store the result in memory.

ALGORITHM

1. Load the first number into register A.
2. Load the second number into register B.
3. Move the contents of A to a temporary register C.
4. Move the contents of B to A.
5. Move the contents of C to B.
6. Store the swapped numbers in memory.
7. Stop the program.

CODE:

```
MVI A,25H
MVI B,35H
MOV C,A
MOV A,B
MOV B,C
STA 8050H
MOV A,B
STA 8051H
HLT
```

OUTPUT:

FileResetAssemblerDebugHelp

Registers

A28

BC2800

DE0000

HL0000

PSW0000

PC420F

SPFFFF

Int-Reg00

Flag

S1

Z0

AC0

P1

C1

Decimal - Hex Conversion

Decimal

0

Hex

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

2501

-

+

19

Update Memory

Load me at

1LDA2500

2MOV B, A

3

4LDA2501

5STA2502

6

7MOV A, B

8STA2503

9HLT

DataStackKeyPadMemoryI/O Ports

Start2500

OK

Address (Hex)AddressData

09C4250040

09C5250125

09C6250225

09C7250340

09C825040

09C925050

09CA25060

09CB25070

09CC25080

09CD25090

09CE25100

09CF25110

09D025120

09D125130

Line NoAssembler Message

0

Program assembled successfully

Simulator: Idle

Search

ENG

IN

11:44

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